



UNIVERSITAT POLITÈCNICA DE CATALUNYA
BARCELONATECH

Centre de la Imatge i la Tecnologia Multimèdia

The Protocol Stack

2.4 Putting it all together: Wireshark

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Xarxes i Jocs On-line – 2025-2026
Grau en Disseny i Desenvolupament de Videojocs

1. Introduction
2. The protocol stack (3 sessions)
 - Introduction and upper layers
 - Lower layers: network, link, physical
 - Other relevant protocols
 - **Putting it all together: Wireshark**
3. Impact on programming of on-line games
4. Other stories

Remember from last class



- We covered all layers of the protocol stack, with some basic protocols and some other relevant ones, like:
 - ICMP
 - NAT
 - TLS
 - RTP
 - HTTP
 - DNS
- These are important to have a hassle-free experience in gaming, from boot-up to setting up the connection.

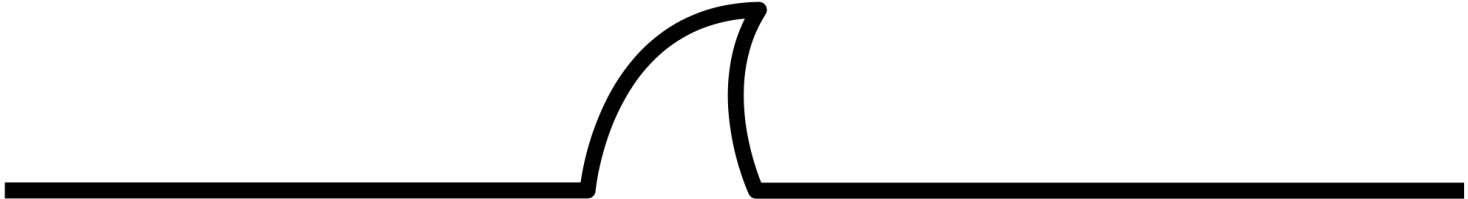


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Packet Sniffers: Wireshark

- Formerly known as Ethereal, Wireshark is a **packet sniffer** or **packet analyzer**.
- It is attached to a network interface of your equipment and, given the necessary permissions, captures all the packets that go through it

The Wireshark logo consists of a thick horizontal line with a curved, shark-fin-like shape rising from its center.

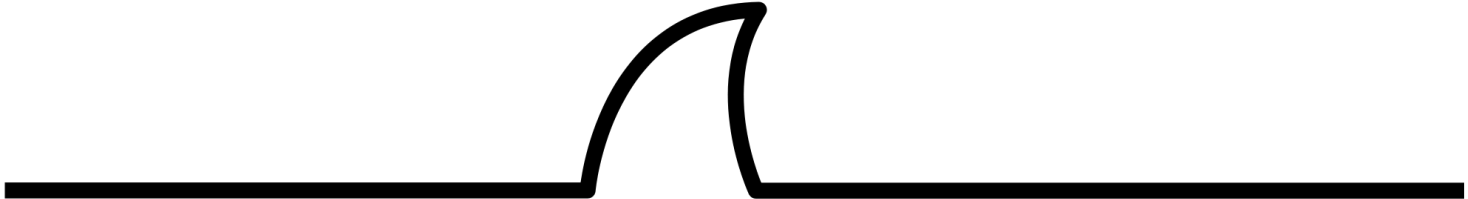
WIRESHARK

Wireshark: uses



CITM

- Development and debugging of protocols
- Debugging of client-server communications
- Testing of systems
- Traffic analysis
- Utilization monitoring
- ...



WIRESHARK

The Wireshark logo consists of a stylized shark fin above a horizontal line, with the word 'WIRESHARK' in a bold, sans-serif font below it.

Inspection tools are the key!

- Deep inspection and analysis tools will help you identify the best optimizations
- Think about the kind of tools you use for CPU performance optimization



Wireshark: uses



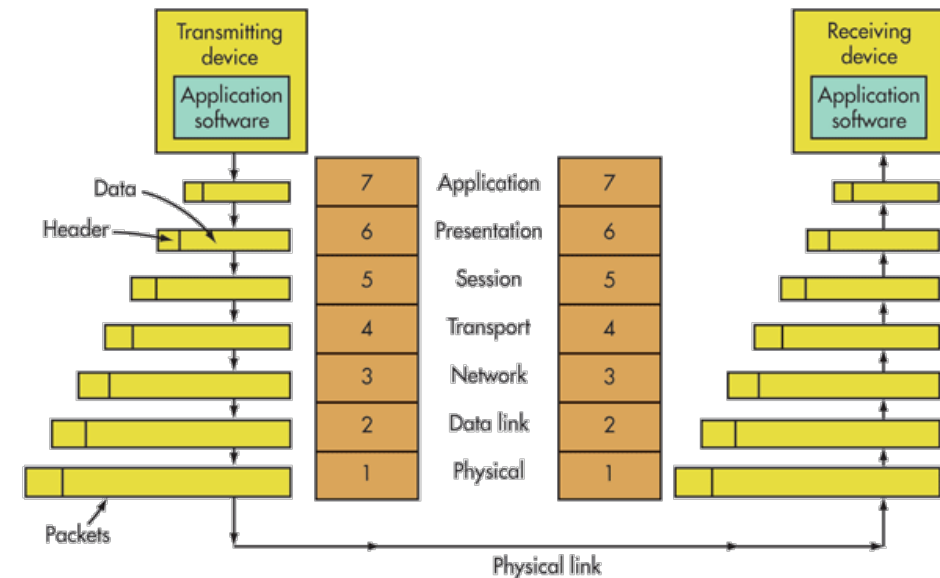
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- The network card in your computer is at the data link level (Ethernet or WiFi)
- Therefore, all packets captured by wireshark can tell you information about the layers...



Wireshark: uses

- The network card in your computer is at the data link level (Ethernet or WiFi)
- Therefore, all packets captured by wireshark can tell you information about the layers **FROM APPLICATION DOWN TO DATA LINK LAYER**



Packet analyzers

- Although there are many alternatives, the operation principles are very similar.
- We will learn to use Wireshark.

Notable packet analyzers [\[edit \]](#)

For a more comprehensive list, see [Comparison of packet analyzers](#).

- | | | |
|--|---|---|
| • Capsa Network Analyzer | • Kismet | • snoop |
| • Charles Web Debugging Proxy | • Lanmeter | • tcpdump |
| • Carnivore (FBI) | • Microsoft Network Monitor | • Observer Analyzer |
| • CommView | • NarusInsight | • Wireshark (formerly known as Ethereal) |
| • dSniff | • NetScout Systems nGenius Infinistream | • Xplico Open source Network Forensic Analysis Tool |
| • EndaceProbe Analytics Platform by Endace | • ngrep , Network Grep | |
| • ettercap | • OmniPeek , Omnipliance by Savvius | |
| • Fiddler | • SkyGrabber | |



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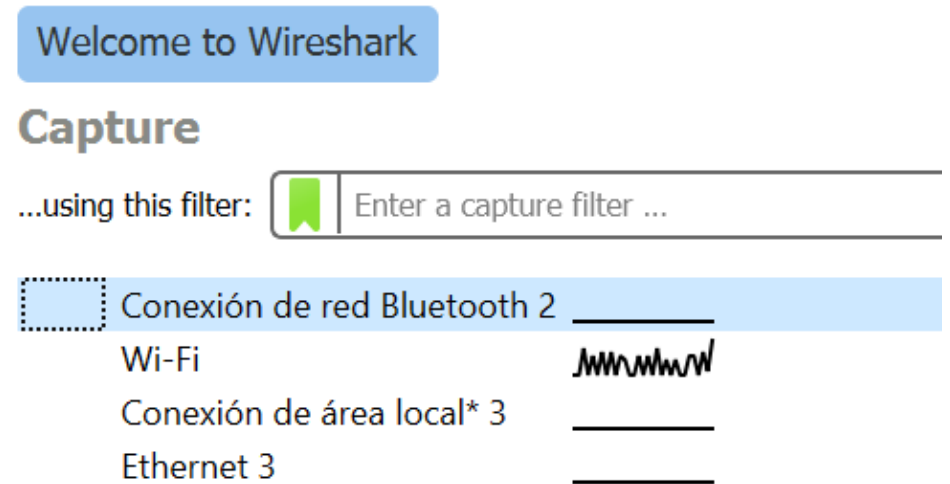
Learn to use Wireshark: Capture and filter

Learning to use Wireshark (I)

1. Open wireshark. You can see the available interfaces and their activity.

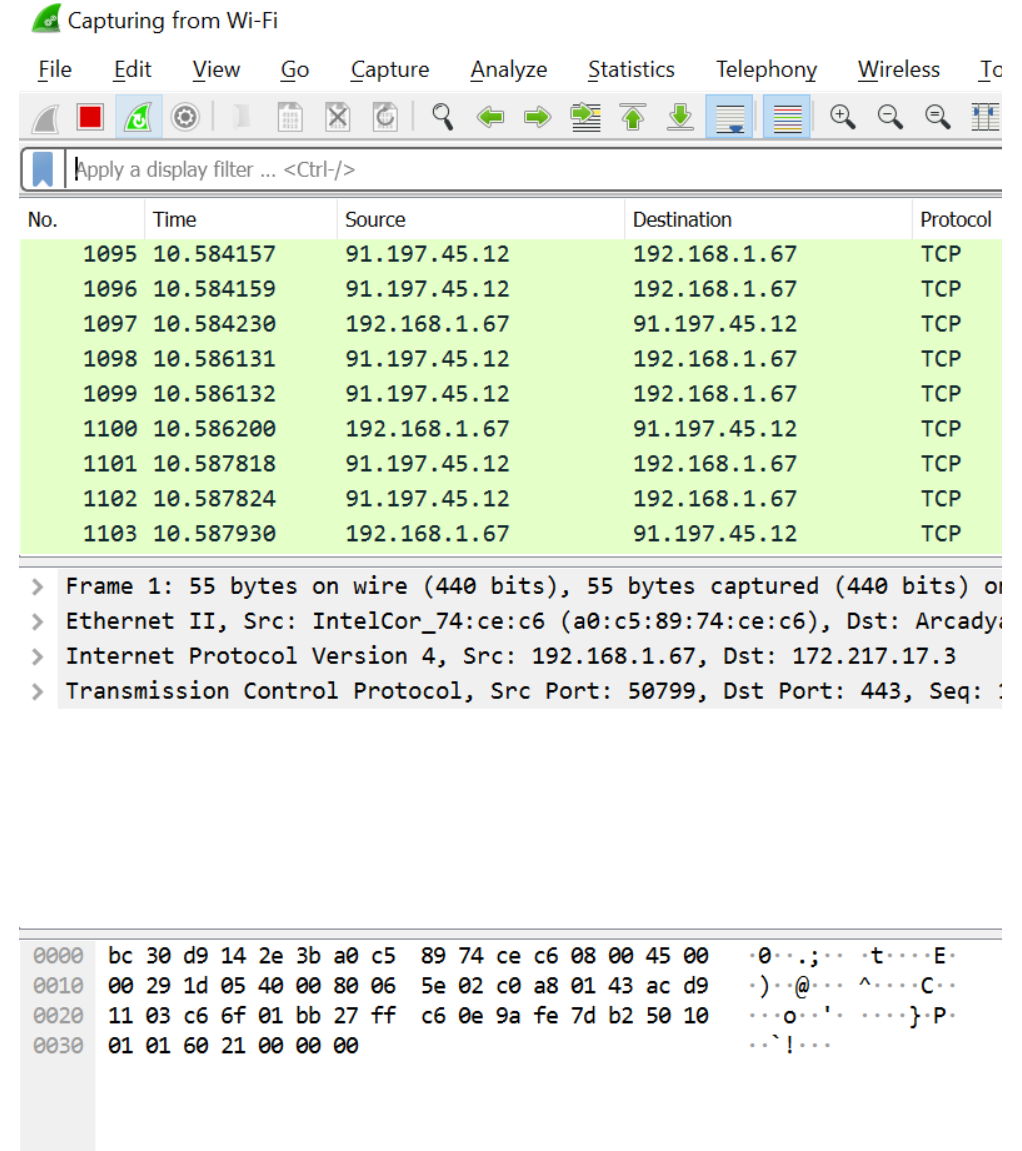
You can choose one and press enter (we will see the filter options later)

Which one has more activity?



Learning to use Wireshark (II)

1. Open wireshark.
2. **Start the capture.**
You will see three parts;
(a) on top you have the list of captured packets,
(b) in the middle you have the parsed information about a selected packet,
(c) at the bottom you have the raw contents of the packet.



Capturing from Wi-Fi

File Edit View Go Capture Analyze Statistics Telephony Wireless Tools

Apply a display filter ... <Ctrl-/>

No.	Time	Source	Destination	Protocol
1095	10.584157	91.197.45.12	192.168.1.67	TCP
1096	10.584159	91.197.45.12	192.168.1.67	TCP
1097	10.584230	192.168.1.67	91.197.45.12	TCP
1098	10.586131	91.197.45.12	192.168.1.67	TCP
1099	10.586132	91.197.45.12	192.168.1.67	TCP
1100	10.586200	192.168.1.67	91.197.45.12	TCP
1101	10.587818	91.197.45.12	192.168.1.67	TCP
1102	10.587824	91.197.45.12	192.168.1.67	TCP
1103	10.587930	192.168.1.67	91.197.45.12	TCP

> Frame 1: 55 bytes on wire (440 bits), 55 bytes captured (440 bits) on interface 0

> Ethernet II, Src: IntelCor_74:ce:c6 (a0:c5:89:74:ce:c6), Dst: Arcadya_08:00:27:00:00:00

> Internet Protocol Version 4, Src: 192.168.1.67, Dst: 172.217.17.3

> Transmission Control Protocol, Src Port: 50799, Dst Port: 443, Seq: 3092128000

Offset	Hex	ASCII
0000	bc 30 d9 14 2e 3b a0 c5 89 74 ce c6 08 00 45 00	.0...;...t...E..
0010	00 29 1d 05 40 00 80 06 5e 02 c0 a8 01 43 ac d9	..@...^...C...)
0020	11 03 c6 6f 01 bb 27 ff c6 0e 9a fe 7d b2 50 10	...o...'...}..P..
0030	01 01 60 21 00 00 00	...!...

Learning to use Wireshark (III)

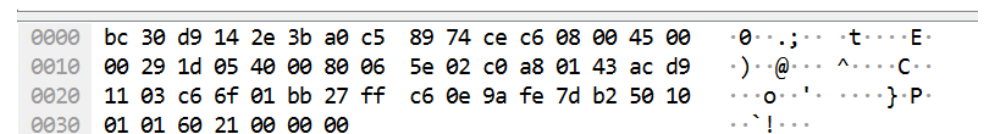
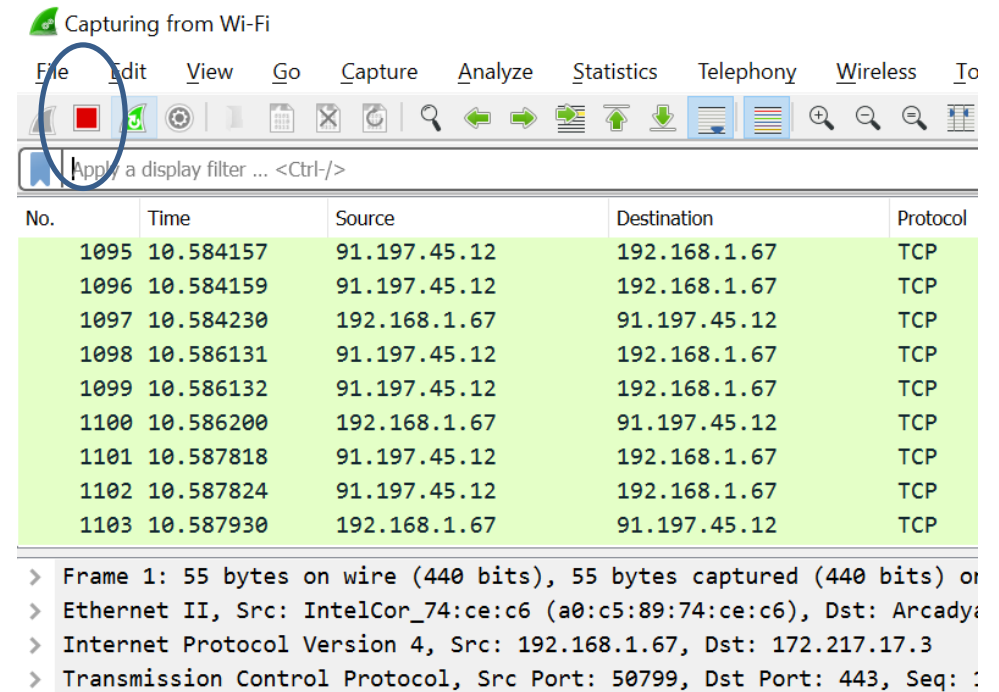


1. Open wireshark.

2. Start the capture.

3. **Stop the capture.**

Otherwise, it is impossible to see anything because we are continuously connected.

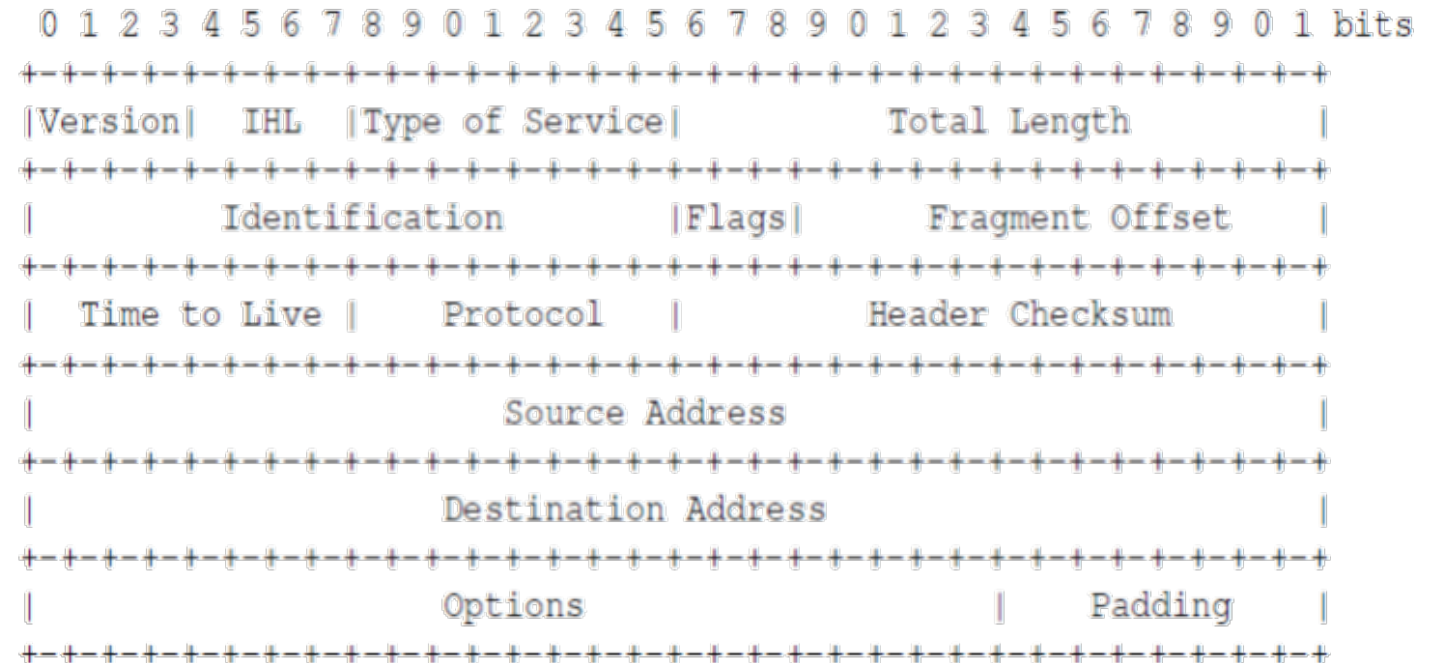


Example



- Navigate a bit.
- Then, select any TCP/IP packet and observe the fields and values of the

IP header



Example (II)

- Navigate a bit.
- Then, select any TCP/IP packet and observe the fields and values of the

TCP header

```
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|           Source Port           |           Destination Port           |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                               Sequence Number                               |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                               Acknowledgment Number                               |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
| Header |           | U | A | P | R | S | F |           |
| length | Reserved  | R | C | S | S | Y | I |   Advertised window   |
|         |           | G | K | H | T | N | N |           |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|           Checksum           |           Urgent Pointer           |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
|                               Options                               |   Padding   |
+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+---+
```


Example (II)

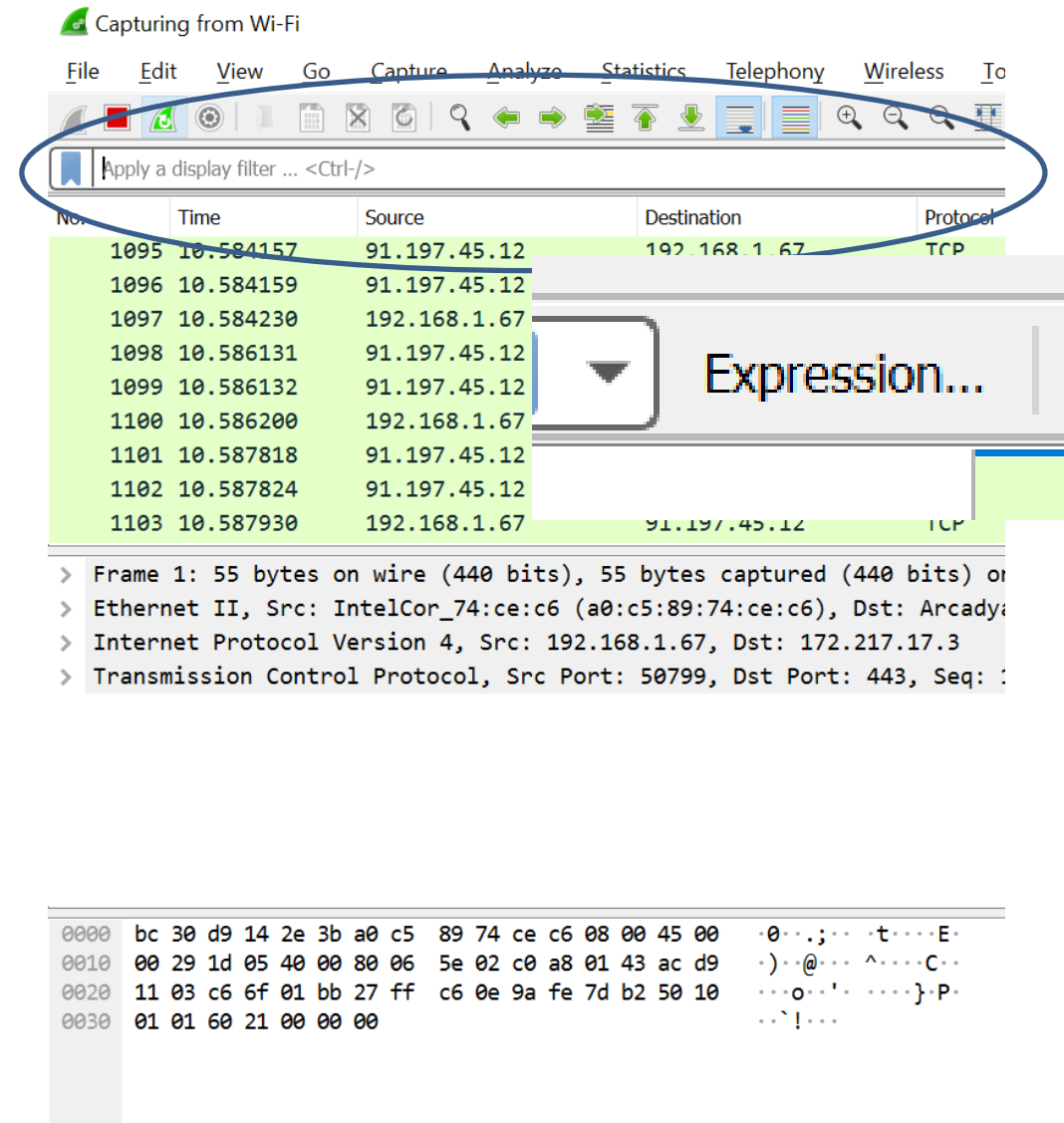
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- Then, select any TCP/IP packet and observe the fields and values of the

TCP header

```
0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1 2 3 4 5 6 7 8 9 0 1
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
|                               Source Port                          |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
|                               Sequence Number                      |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
|                               Acknowledgment Number               |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
| Header |                   |U|A|P|R|S|F|                          |
| length| Reserved  |R|C|S|S|Y|I|      Advertised window           |
|         |                   |G|K|H|T|N|N|                          |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
|                               Checksum                             |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
|                               Options                               |
+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+--+
```

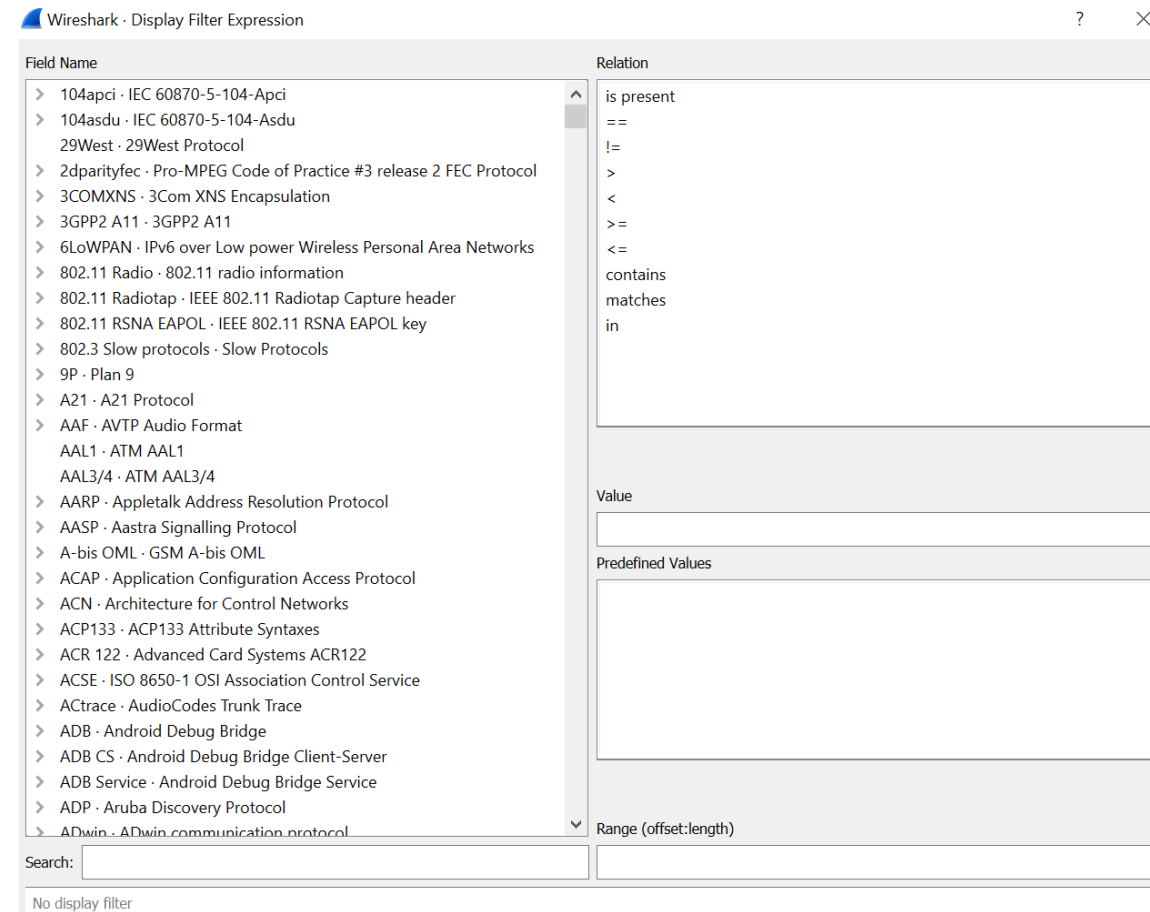
Learning to use Wireshark (IV)

1. Open wireshark.
2. Start the capture.
3. Stop the capture.
- 4. Filter the captured packets.**
If you see too much information, you can use filter to see what you are looking for.



Learning to use Wireshark (IV)

1. Open wireshark.
2. Start the capture.
3. Stop the capture.
- 4. Filter the captured packets.**
If you see too much information, you can use filter to see what you are looking for.



Exercise

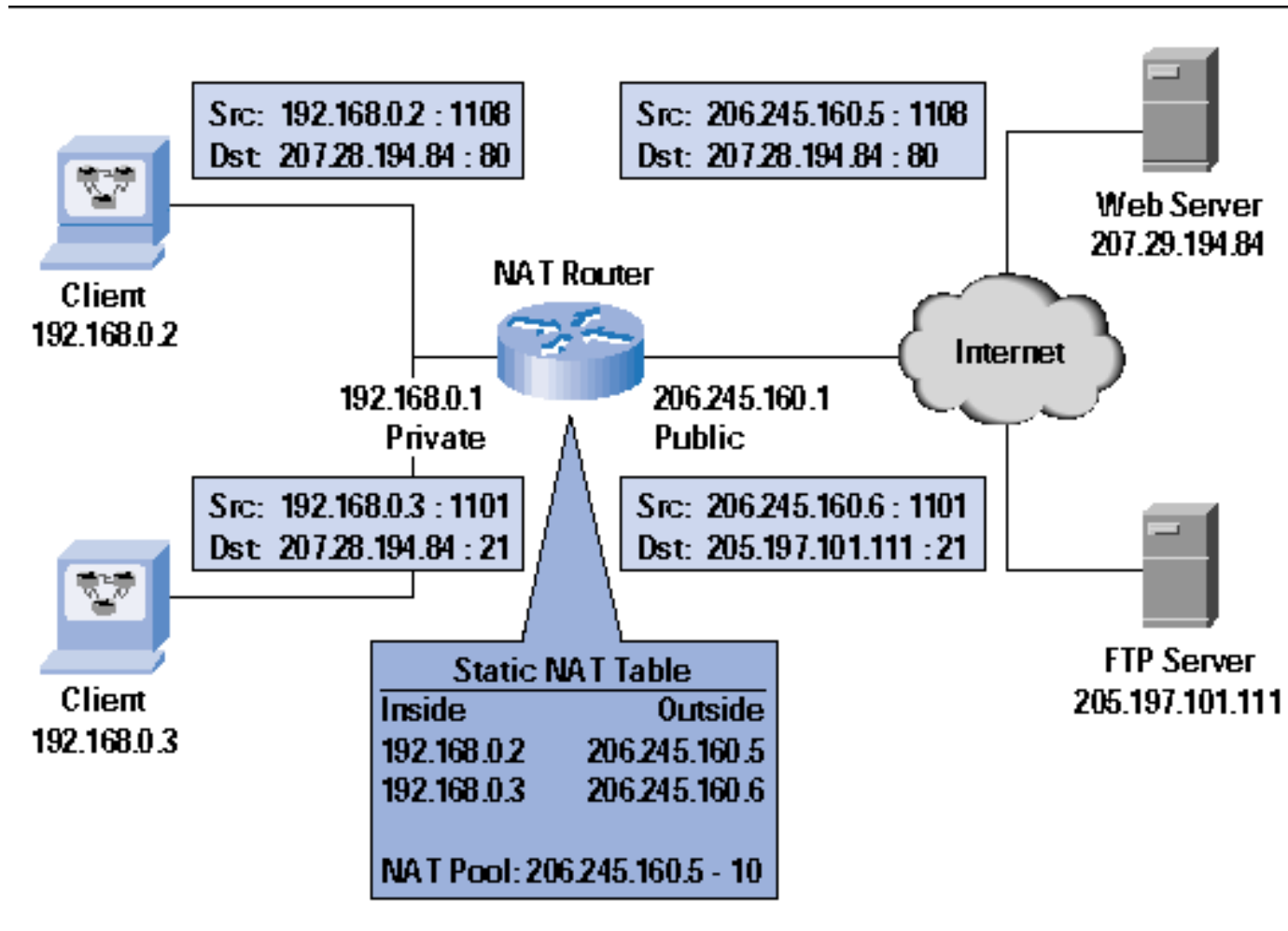


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- A. Find out your public IP address and your private IP address (ipconfig?)
- B. Create a filter that only shows packets coming from or arriving to your public IP address.
What happens?
- C. Then, change that for the private IP address.
What happens?



Exercise: Remember NAT...



Exercise



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- Start a new session and create a display filter that only shows ICMP messages.
- Then make a ping to any address and observe the traffic.



Exercise



CITM

- Close everything you can to avoid unnecessary traffic.
- Then open a new session and access:
<http://personals.ac.upc.edu/acabello/>
<http://httpforever.com/> or <http://www.faqs.org/faqs/>
- As soon as the page has charged, stop the capture.
- **Reason about DNS traffic; HTTP traffic; others?**
- For some of the packets at the upper layers, check:
 - What is the size of the payload (data)?
 - What is the size of the headers?
 - Calculate the header to application data ratio



Learning to use Wireshark (V)



CITM

- You can see the statistics of the type of traffic you have captured in a trace, divided in the different layers of the protocol stack. You can do it selecting:

Statistics > Protocol Hierarchy

Wireshark · Protocol Hierarchy Statistics · Wi-Fi (port 22)

Protocol	Percent Packets	Pack	Percent Bytes	Bytes	Bits/s
▼ Frame	100.0	3	100.0	198	175
▼ Ethernet	100.0	3	21.2	42	37
▼ Internet Protocol Version 4	100.0	3	30.3	60	53
Transmission Control Protocol	100.0	3	48.5	96	85

Exercise



CITM

- Navigate through the internet, make pings, make speed tests, download files, watch videos in YouTube... During five minutes.
- Then check the statistics. Identify as many protocols as you can (the more layers, the better) and reason about the type of traffic you find.
 - IP? ICMP? TCP? UDP? TLS? DTLS? QUIC? PNG? DNS? HTTP? RTP?
- Stop simulation, save the trace in your computer.
Check the size. Is this affordable?





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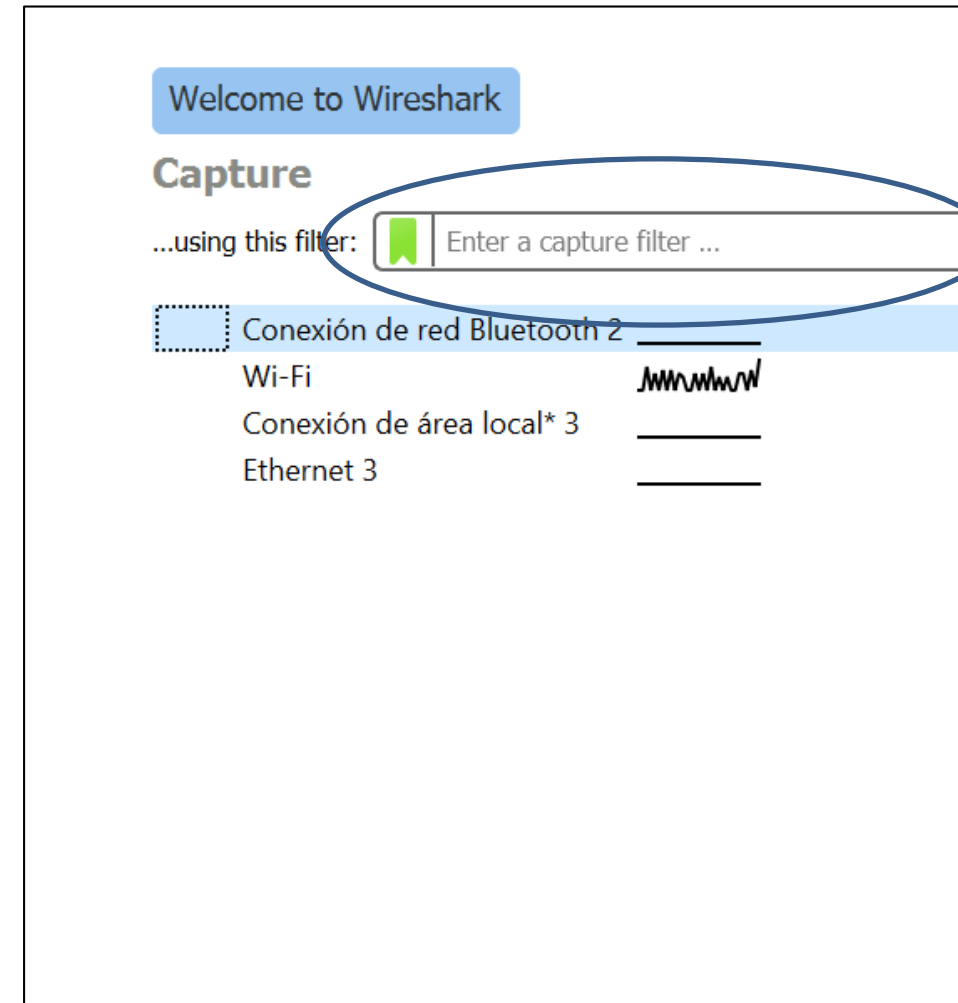
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Learn to use Wireshark: Filter and capture

Learning to use Wireshark (VI)

- Restart wireshark.
- **Choose a capture filter.** If you know beforehand what you want to look for, you can introduce a capture filter (different from a display filter) to avoid garbage

<https://wiki.wireshark.org/CaptureFilters>



Exercise – Capture Filters



CITM

- Create a capture filter that would show only messages related to the DNS protocol. **How would you force DNS traffic to test this?**



Exercise



CITM

- Use the code that you programmed in the lab sessions to create a TCP or UDP socket, you choose.
- Create a capture filter to capture the traffic that you are going to generate with your code (using the protocol? The port? Your address? The destination address?)
- Send some message.
- Check the size of the messages and of the headers
- Try to send something larger than 2KB, and check what happens. I dare you to try to see fragmentation





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